Abdullah Naveed

abdullahnaveed102@gmail.com | abdullahnaveed102.github.io | linkedin.com/in/abdullah-naveed-3babbb22b

Education

University of Florida — Expected May 2027
Ph.D. Candidate in Computer Engineering | GPA: 3.93/4.00 — *Gainesville, FL*

University of Iowa — Aug 2023 – Aug 2025
Master of Computer Science | GPA: 3.99/4.00 — *Iowa City, IA*

Lahore University of Management Sciences — Aug 2019 – May 2023
B.S. in Computer Science | GPA: 3.61/4.00 — *Lahore, Pakistan*

Experience

Argonne National Laboratory — Jun 2025 – Present
Visiting Graduate Researcher — *Lemont, IL*

- Designed an end-to-end **FPGA/CPU scientific data-compression framework**, achieving up to **19.8× cycle speedup** over matched software and **46–65% lower end-to-end workflow time**.
- Developed a **modular Chisel/RTL compression library** for composing system-optimized pipelines, supporting LZ4, Snappy, StreamVByte, SZ3, SZp, SZx, zfp, and extensible new designs.

University of Florida — Aug 2025 – Present
Graduate Research Assistant, Dependable Systems Lab — *Gainesville, FL*

- Developed **TRACE**, a runtime detector for silent data corruption during large language model (LLM) inference, achieving **95.3% average detection coverage** across 15 LLMs and 3 datasets with **0.56% runtime overhead**; paper accepted at SC26.
- Characterize **LLM resilience to transient hardware faults** during inference and pretraining, and in models fine-tuned with **LoRA, QLoRA, LoRA-FA, and DoRA**, analyzing sensitivity to fault intensity and parameter corruption.

University of Iowa — Aug 2023 – Aug 2025
Graduate Research Assistant — *Iowa City, IA*

- Developed an **LLVM-based silent-error detector** for scientific and HPC applications, achieving **86.19% average detection coverage at 4.90% runtime overhead** across 15 benchmarks.
- Compared **compiler- and runtime-based error-detection techniques** across scientific workloads, analyzing their behavior under different fault-propagation patterns and quantifying trade-offs between **detection coverage and runtime overhead**.

Technical Projects

- **DebloatBench / BloatProfiler** — developed an extensible **Python and Docker benchmarking framework** for container debloaters, measuring attack-surface reduction, vulnerability mitigation, image-size reduction, and functional correctness; **published at IEEE SecDev 2023**.
- **Multimodal Fake News Classification** — developed and benchmarked text–image classifiers using **BERT, CNN vision models, and multimodal fusion**, comparing approaches with accuracy, precision, recall, and F1.
- **LUMS Mobile App** — developed a cross-platform student application using **React Native, JavaScript, and Firebase**, integrating deadline management, GPA tracking, communication, and real-time cloud-backed data.

Publications

- *Low-Overhead Silent Error Detection for Large Language Model Inference at Scale.* SC26. **Accepted; to appear.**
- *Z-SPDK: Accelerating Lossy Compressed I/O on NVMe-SSDs.* SC26. **Accepted; to appear.**
- *Evaluating Container Debloaters.* IEEE SecDev, 2023.

Technical Skills

Programming: Python, C, C++, JavaScript, SQL, Bash, Git
Machine Learning / Data: PyTorch, Hugging Face Transformers, TensorFlow/Keras, scikit-learn, OpenCV, Pandas
Systems / HPC: LLVM, Linux, Docker, MPI, OpenMP, Slurm, fault injection, parallel computing
Hardware: Chisel, Verilog/RTL, RISC-V, Chipyard, Verilator, Vivado
Software Development: React, React Native, Firebase, HTML/CSS